

Hieu Pham

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Profile

I'm a career changer with 5+ YOE in financial data analysis

Skills

Programming Languages C, C++, Bash, TypeScript, Go

Development Tools Git, Linux, Make, GDB, Docker, Docker Compose, Neovim

Projects

Web Server - HTTP Server from scratch

- Engineered a low-level C++ HTTP server via POSIX sockets, handling TCP connections and serving static HTML, CSS, Javascript, and image files
- Delivered full GET, POST, and DELETE support with spec-compliant status codes and response header
- Enabled concurrent browser and client support via poll-based non-blocking I/O without threads

Interception - Dockerized Full Stack Web App

- Architected a multi-container web application using **Docker** and **Docker Compose**, running services in isolated, lightweight environments
- Configured **Nginx** as a reverse proxy with custom SSL/TLS encryption (HTTPS), managing secure traffic routing, port bindings, and optimized request handling for backend services
- Integrated a custom **WordPress** frontend connected to a dedicated **MariaDB** database with persistent volume management

Angry Cube - A pseudo 3D game

- Implemented a 2D raycasting graphics engine from scratch in C
- Developed robust 2D linear vector mechanics for player movements and collision detection

SuperShell - A lightweight shell interpreter

- Engineered a POSIX-compliant Unix shell in C with multi-process execution, pipeline routing and I/O redirection using system calls
- Designed a custom memory arena allocator to optimize heap usage and prevent memory fragmentation and leaks
- Built a custom lexer and a parser adhering to standard shell token recognition rules
- Implemented unit testing to ensure the robustness of the code base

Huff - A text compression tool

- Implemented the Huffman algorithm to compress texts

Cfit - An activity visualizer

- Explored a Developer Kit library for Garmin devices
- Analyzed the activity data and created a visualization using Raylib

Dining Philosophers - Concurrency and Parallelism

- Implemented a multi-threaded and multi-process program to solve the classic dining philosopher problem using POSIX pthread and semaphore

Professional Experience _____

Senior Data Analyst (*Financial Recovery Technologies*) 2019-2024

- Exercised full accountability and project management skills by overseeing the lifecycle of a class action case
- Crafted PL/SQL scripts to automate calculation of clients’ entitled losses
- Translated highly technical legal and financial data into clear, actionable information for diverse clients

Education _____

Hive Helsinki	Software Engineer
Suffolk University	Master in Business Analytics
Northeastern University	Bachelor in Finance and Accounting Management